

Shear Extension Stabilizer

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Summary

Pass 88 computes the stabilizer of the final finite-adele shear extension

$$0 \rightarrow \mathbb{Q} \rightarrow \mathbb{A}_f \rightarrow \epsilon \rightarrow 0$$

and separates it from the full solid Borel $\mathbb{Q}^\times \ltimes \epsilon$.

Main Claims

- **Strict marked rigidity.** As an object under $C_{\mathbb{Z}} \rightarrow C_{\mathbb{Q}}$, the extension has trivial automorphism group. The marked integral unit forces scalar 1, and Pass 87 removes derived ambiguity for the $\mathbb{Q}$ -kernel.
- **Extension-line stabilizer.** Forgetting the integral marking but preserving the finite-adele Ext line leaves stabilizer $\mathbb{Q}^\times$, acting by nonzero rational scalars.
- **Borel comparison.** The full Borel $\mathbb{Q}^\times \ltimes \epsilon$ is recovered only at the hyperbolic-plane level $H = \epsilon \oplus \mathbb{Q}$, where ϵ is the unipotent shear parameter rather than an endpoint-fixing automorphism of the bare exact row.

Machine Verification

`code/scripts/check-pass88.py` produced `artifacts/reports/pass88-shear-extension-stabilizer-check.json` with overall PASS. It checks rational scalar behavior, finite affine Borel shadows $(\mathbb{Z}/N)^\times \ltimes \mathbb{Z}/N$, singleton strict unit stabilizers, and absence of residual derived automorphisms after the torsion-boundary decoration rule.

Next Use

Pass 89 should consolidate Passes 80-88 as a Borel-torsor / extension-class theorem for the Rosser phantom and state the bridge back to the APS/Rosser unit-torsor line.